

Queryable Canonical Surfaces from Sparse Driving Observations

WorldSim V7.3 Research Draft

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Abstract

We study reconstruction of a reusable, metric surface for each rigid actor from sparse multi-time observations. A single canonical surface must explain measured points and remain consistent when queried from different poses. This task exposes a gap between geometric coverage and first-surface correctness: a surface can lie close to a measured endpoint while a different part intersects its ray too early. We formulate queryable canonical surfaces and develop witness-constrained charts, separating surface placement, support domain, and observation responsibility. An oracle ladder and exact triangle forensics on 75 development actors show that well-shaped, observation-supported patches can still create incorrect first intersections. Across seven saved representations, none of the early-hit triangles exceeds the stated sliver threshold. The completed joint r7 pathway improves correct first intersections by 10.54 percentage points over its LiDAR control, but preservation of free-space consistency is not established. We derive a sparse first-intersection adjoint and an incidence-normalized orthogonal update that protects fixed first-hit witnesses during completion. The new operator passes numerical checks; learned support domains and end-to-end orthogonal training remain unvalidated. Completed experiments, checked prototypes, and proposed components are reported separately. The manuscript advances a bounded reconstruction task and testable method, without claiming a finished simulator or established overall superiority.

1. Introduction

A sparse scan records where a beam terminates, but it also constrains the space before that return. Recovering a reusable surface therefore asks more than placing points near observations. The reconstructed object must give coherent answers when the same surface is intersected from another time or viewpoint. A completion patch can improve endpoint coverage while obstructing a beam that should reach a different patch. Optimizing one aggregate distance can conceal this failure.

We define the task as *recovering a queryable canonical*

rigid surface from sparse observations. The output is an actor surface in metric canonical coordinates, shared across all query rays and rigid placements. Known identities, calibration and trajectories are inputs. Surface reconstruction is the independent deliverable; first-intersection queries test that deliverable. Learning a complete LiDAR sensor, appearance model, motion policy, or closed-loop world simulator is outside the task.

This reformulation follows a useful lesson from DVGT [1]: an output representation should express the geometric object required by the task. DVGT predicts metric point maps in a common ego frame. Our output instead must be independent of the image raster and reusable under an actor’s rigid motion. The contribution cannot consist merely of renaming a point map, adding a small head, or verifying a coordinate identity. It must supply a shared surface, an observation-consistent learning mechanism, and evidence that the surface answers the intended queries.

Before proposing that mechanism, we ask what the extra surface actually looks like. A thin stretched triangle suggests shape regularization; an unsupported component suggests selective pruning; a well-shaped but overextended patch suggests a support-boundary or visibility problem. These explanations require different mathematics. We trace the *actual first-hit triangle*, render the saved geometry in Blender, and intersect it with a plane containing the sensor ray. We then relax the evaluation in stages: correct first hit, any correct intersection, and any nearby surface. This locates what can be recovered by choosing an existing intersection and what still requires changing geometry.

The resulting evidence motivates *witness-constrained canonical charts*. Each chart couples an embedding with a physical support domain and sparse observation witnesses. For fixed visibility, first-hit depth correction lies in the row space of a sparse geometric Jacobian; completion can move in its null space. Multiplying by the incidence term yields a constraint without the grazing-angle denominator. A separately represented chart boundary can express partial support without distorting otherwise well-shaped triangles. These are method-level hypotheses, not consequences of a stronger backbone. We implement and check the sparse operator, while explicitly marking the remaining training and

boundary-learning work.

The manuscript contributes (i) a bounded surface-reconstruction formulation with a shared query contract; (ii) original failure morphology, oracle and pruning evidence linking measurements to responsible triangles; and (iii) a concrete surface-domain formulation and sparse witness-orthogonal operator. Current r3–r7 experiments test the existing chart and first-surface pathway. They do not validate every component of the proposed representation.

2. Related Work and Design Rationale

Geometry representations. VGGT [2] provides multi-view geometry features; DVG [1] adapts the output coordinate system and aggregation to driving. AtlasNet [3] establishes chart-based surface generation. AdaPoinTr [4] predicts completed point sets. We reuse these established ideas rather than claim canonical coordinates, charts, or point completion as new. The proposed distinction is a *partially observed, physically queryable surface with explicit witness responsibility*, evaluated on the same exported geometry across rays. Our rendered AdaPoinTr baseline is the full task-fine-tuned model followed by the documented PCA patch conversion; its conversion artifacts cannot all be attributed to native point prediction.

Representation and query operator. LiDAR-RT [5] couples Gaussian primitives carrying intensity and ray-drop properties with a differentiable ray-tracing pipeline. Its operator and representation jointly serve sensor re-simulation. We adopt that level of coupling as a design criterion, while solving a narrower surface task: our query returns the first geometric intersection and its owner, without learned transparency or a sensor-return head. LiDAR-RT is a methodological reference here, not a reproduced numerical baseline.

Diagnose before calibrating. FoundationGeo [6] uses progressively local scale alignment to expose residual geometry error, then separates scale and direction with a tangent-space parameterization. Our oracle ladder similarly localizes an error before selecting an intervention. Its percentages describe support availability, not a model allowed to choose a target-dependent return. We also retain non-dominating benchmark outcomes: FoundationGeo’s reported metric AbsRel is 10.2 versus MoGe-2’s 7.33 on NYUv2 and 17.8 versus 10.4 on ETH3D; lower is better. An average improvement does not imply superiority on each domain.

Visibility and constrained updates. Point2Mesh [7] uses a mesh self-prior and beam-gap attraction. Distance-based attraction does not by itself select the first visible sur-

face. Differentiable rendering distinguishes smooth interior derivatives from visibility discontinuities [8, 9]. Gradient projection is also established [10, 11]. Our proposed use is specifically tied to the sparse first-intersection Jacobian and incidence normalization of a shared canonical surface. We do not claim to invent null-space projection or to solve silhouette derivatives through an interior formula.

3. Queryable Canonical Surface Reconstruction

Let $\mathcal{O}_i$ contain build-side camera frames, LiDAR endpoints, beam origins and directions associated with rigid actor i . A known transform $T_i(t) = (R_i(t), t_i(t))$ maps canonical actor coordinates to world coordinates. Sensor and ego calibration map each endpoint and its origin into this frame. The model predicts

$$f_\theta(\mathcal{O}_i) = \mathcal{R}_i, \quad \mathcal{S}_i = \bigcup_{k=1}^K \psi_{ik}(\Omega_{ik}) \subset \mathbb{R}^3. \quad (1)$$

Here $\psi_k : \Omega_k \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a chart embedding. The stored representation also records source provenance and a sparse witness index $\mathcal{W}_i$. Geometry is shared: no surface is constructed separately for an evaluation ray.

For a world-space ray (o, d) , the geometric query is

$$o_i = R_i^\top(o - t_i), \quad d_i = R_i^\top d, \quad (2)$$

$$Q(\mathcal{R}_i, o, d, t) = \inf\{s \geq 0 : o_i + s d_i \in \mathcal{S}_i\}. \quad (3)$$

The value is $+\infty$ when there is no intersection. Closest-point queries and mesh export use the same surface. A common rigid change of coordinates preserves the range in Eq. 3, since rotations preserve lengths. This identity validates reuse of coordinates; it does not prove reconstruction accuracy.

Observation semantics. A measured return (o_r, d_r, r_r) gives endpoint $x_r = o_r + r_r d_r$ and evidence about the pre-hit beam interval. The region behind the first return is unobserved. It is not a negative surface target. Sparse endpoints also do not provide the complete ground-truth surface, so our main distance is directed target-to-surface distance, not unrestricted symmetric Chamfer accuracy. Missing predictions and actors without owned heldout returns remain in the cohort; an undefined physical metric is not a successful zero.

What is new in the task. The required output is a reusable partial surface constrained by multiple observations, rather than independent per-view depth or a per-ray return predictor. Querying checks surface coherence across views. There may be many surfaces consistent with sparse

data; the task does not promise unique hidden geometry. The intended success criterion is improved measured coverage together with credible first-surface and free-space consistency at a fixed observation budget.

4. Witness-Constrained Canonical Charts

4.1. Implemented Embedding and First-Surface Learning

The current joint path adapts the full native VGGT DPT depth head (32,654,562 parameters), reads four feature levels, and uses local spatial queries to generate actor charts. Only the fully frozen aggregation prefix is cached. Geometry gradients reach the trainable DPT and its sampled features. Native seeds, feature reading and build-depth auxiliary supervision together define the joint pathway; r7 versus r6 does not isolate a single feature layer.

The open-chart models use 64 charts, each a 4×4 vertex grid, giving 1,024 vertices and 1,152 triangles. A chart has a center, tangent frame, normal and height offsets. Its embedding has the form

$$\psi_k(u) = c_k + U_k u + n_k h_k(u), \quad U_k^\top U_k = I, \quad U_k^\top n_k = 0. \quad (4)$$

In r3–r7 the chart domain is a fixed square for each actor’s configured scale. This structural distinction matters: a regular local triangulation can still cover the wrong area.

For endpoint residual $\delta = \psi_k(u) - x_r$, define $P_\parallel = d_r d_r^\top$ and $P_\perp = I - P_\parallel$. The constructive attraction used in r4 is

$$L_{\text{att}}(r) = \min_{k, u \in \Omega_k} \|\alpha P_\perp \delta + P_\parallel \delta\|_2, \quad \alpha = 20/3. \quad (5)$$

The ratio reflects the current 0.03 m lateral and 0.2 m range scales, not a calibrated noise model. The nearest point is solved on actual triangles in the anisotropic metric. Discrete selection is detached; the envelope theorem gives the selected minimum’s derivative away from switches. Merely detaching an arbitrary closest point would not justify other downstream derivatives.

Attraction can select a correct rear surface while ignoring an incorrect front one. The r6/r7 objective therefore uses

$$L_{\text{first}}(r) = \begin{cases} |Q_r - r_r|, & Q_r < \infty, \\ L_{\text{att}}(r), & Q_r = \infty. \end{cases} \quad (6)$$

The total loss adds coverage, 0.5 times finite-beam free loss, 0.05 times box-envelope loss, and unit-weight Eq. 6; r7 also includes the native build-depth auxiliary term. No opacity or target-dependent return selector changes the deployed surface. The miss branch moves existing geometry; it cannot create a surface from an empty mesh.

4.2. Surface Domain as a First-Class Unknown

The proposed representation is

$$\mathcal{R}_i = (\{c_k, U_k, n_k, h_k, b_k\}_{k=1}^K, \mathcal{W}_i), \quad \Omega_k = \{u \in D_k : b_k(u) \geq 0\}. \quad (7)$$

The scalar b_k describes an intrinsic *surface boundary*, not opacity, ray-drop probability, or occupied thickness. Its zero contour clips chart triangles into a single exportable geometric surface. The same clipped mesh is used for supervision and all queries. A sparse witness records an observation ID, supporting face/chart, barycentric location, and build-side measurement role. Occluded or unsupported regions remain unknown rather than receiving arbitrary free labels.

This separates *where a surface lies* (ψ) from *where that surface exists within the chart* (Ω). Moving a square to satisfy one measurement can obstruct another ray; changing an intrinsic boundary can remove the obstructing extension while preserving supported locations. Regularizing aspect ratio cannot express that change when all triangles already have good shape.

Boundary learning must preserve measured support. A candidate objective minimizes reconstruction loss subject to reliable build witnesses retaining endpoint support and already-correct first returns:

$$\text{dist}(x_r, \mathcal{S}_i) \leq \epsilon_g, \quad |Q_r - r_r| \leq \epsilon_r \quad (r \in \mathcal{W}_i^+). \quad (8)$$

The pre-hit observations constrain free space only on observed beams. A boundary cannot escape its loss by erasing a positive witness. Conflicting measurements may make Eq. 8 infeasible; explicit slack and reported violations are required, not silent witness deletion. Small unsupported components are not automatically false. This domain-learning component is a concrete proposed method, not a result of the completed r7 run.

4.3. First-Hit Orthogonal Surface Updates

Consider a non-grazing first intersection inside triangle (v_0, v_1, v_2) with barycentric weights β_j , unit normal n , and fixed ray d . Differentiating $o + td = \sum_j \beta_j v_j$ and projecting onto n eliminates tangential barycentric changes:

$$\delta t = \frac{\sum_{j=0}^2 \beta_j n^\top \delta v_j}{n^\top d}. \quad (9)$$

The ordinary depth Jacobian A has only nine scalar entries per ray, but its entries grow near grazing incidence. Define the normalized constraint row

$$B_r = (n_r^\top d_r) A_r, \quad B_r \delta V = \sum_j \beta_{rj} n_r^\top \delta v_j. \quad (10)$$

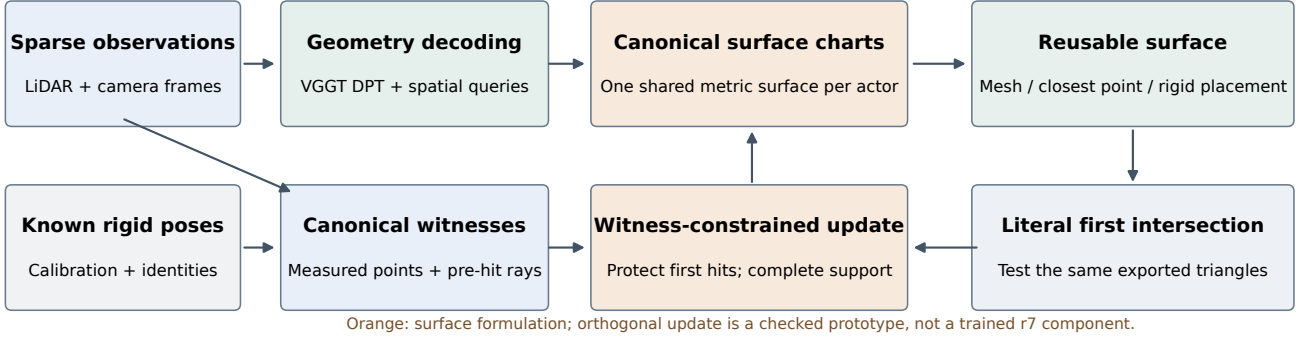


Figure 1. Architecture components and data flow. A single canonical surface is constructed from sparse observations and known rigid transforms. The upper path is implemented in V7.3. The lower witness-orthogonal update is a numerically checked prototype developed in this revision; learned support-domain boundaries are proposed in Sec. 4.2. The query always intersects the exported triangles.

For $n_r^\top d_r \neq 0$, $\ker B_r = \ker A_r$. Normalization changes conditioning, not the local zero-depth-change subspace. It does not make a grazing intersection identifiable.

Let B stack fixed first-hit witnesses, g be a proposed completion displacement, and $B^\dagger$ the Moore–Penrose pseudoinverse. We split the update into

$$\delta V_{\text{support}} = (I - B^\dagger B)g, \quad (11)$$

$$\delta V_{\text{range}} = B^\dagger c, \quad c_r = (n_r^\top d_r)(r_r - t_r). \quad (12)$$

The two components are orthogonal because the first lies in $\ker B$ and the second in $\text{range}(B^\top)$. When c is feasible, $B\delta V_{\text{range}} = c$; otherwise the residual must be reported. If the current witness is already correct, $c_r = 0$ protects its first-order depth while allowing completion directions that respect all stacked witnesses.

Local guarantee and its boundary. With a fixed active triangle, interior barycentrics and nonzero incidence, Eq. 9 is differentiable. Thus $A\delta V_{\text{support}} = 0$, so the protected depths change only at second order along a sufficiently small step. The condition is geometric and per-observation, unlike cancellation in an averaged task gradient. It provides no protection against a new front triangle entering a ray, a boundary crossing, a visibility tie, inconsistent witnesses, or finite-step topology changes. A trainable integration needs shared-surface recomputation and a trust-region/active-set policy; those have not yet been evaluated.

4.4. Sparse Differentiable Operator and Memory Closure

The new prototype uses a CPU BVH to identify each first-hit face, solves the selected intersection, and implements a custom backward:

$$\frac{\partial L}{\partial v_j} = + \frac{\partial L}{\partial t_r} \frac{\beta_{rj} n_r}{n_r^\top d_r}. \quad (13)$$

Only selected vertex indices, coefficients and validity are saved. The incidence-normalized Bu and $B^\top z$ are gather/scatter operators with a matching custom adjoint. A matrix-free LSMR solve projects a proposed displacement using those operators. Coefficients are fixed for this local update; the prototype is not a fully differentiated optimization layer through the changing witness set.

For V vertices, F triangles and R rays, the BVH/mesh and sparse adjoint require $O(V + F + R)$ storage; the solver requires $O(V + R)$ working vectors. It never retains a dense $R \times F$ backward graph. Runtime includes BVH construction, queries, transfers if applicable, and solver iterations; no GPU-speed claim follows from this CPU implementation. Exact edge/boundary derivatives and domain clipping are discussed in the appendix. Their integration, and comparison with the existing r7 operator, remain necessary before claiming end-to-end method gains.

5. Experiments

5.1. Protocol and Evidence Scope

The population contains 489 rigid actors from 31 nuScenes scenes and 25 logs [12]. Of 414 fit entries, 371 have the required build inputs; of 75 development entries, 67 do. All 51 missing-input entries are retained. The development set contains five logs, 11,886 owned heldout returns, and 23 actors without owned heldout returns. Original build measurements are prediction inputs; full-track fit measurements are supervision. Development observations are never optimizer inputs in the completed models. This is an exposed development set, not an untouched test.

Each actor’s observations are pooled, normalized within the actor, averaged within its log, and then equally averaged across logs. We report six metrics: correct first intersection within ± 0.2 m, early intersection, missing intersection, free intrusion in meters, one-way measured target-

to-surface distance, and surface recall within 0.2 m. The owned-return forensics do not replace free-space evaluation on all original near-box beams, including returns owned by other objects. Confidence intervals are the existing 10,000-resample paired log bootstrap; five logs provide limited independent evidence. An interval crossing zero does not establish equivalence.

5.2. What Does the Incorrect Surface Look Like?

We inspect six saved methods and add the completed r7 with the same protocol, not only favorable examples. For triangle area a and edge lengths ℓ_j , use the scale-invariant shape index

$$q = \frac{\ell_0^2 + \ell_1^2 + \ell_2^2}{4\sqrt{3}a}. \quad (14)$$

An equilateral triangle has $q = 1$; a right isosceles grid triangle has $q = 2/\sqrt{3} \simeq 1.155$. We fix $q > 10$ as a descriptive severe-sliver threshold. No observed early-hit face crosses it. The largest such q is 1.367 across these seven representations; r3, r4, r6 and r7 are near 1.155. The regular geometry of PCA conversion makes this partly structural for AdaPoinTr/native surfaces. It is direct evidence against sliver regularization as the immediate explanation for *these* bad cases, not a rejection of mesh regularization generally.

A component is called build-supported when at least one build endpoint has unsigned distance at most 0.2 m to one of its triangles. Disconnectedness alone is insufficient: the current charts are disconnected by construction. On r4, 29.171 of the 30.797 early percentage points originate on build-supported components; on r6 the corresponding values are 20.919 of 22.981. These ratios are 94.7% and 91.0% of the log-weighted early contribution. A supported component can still extend incorrectly away from its supporting observations. This test does not certify every triangle or prove a unique error cause.

The slices reveal regular planar fragments and overlapping chart extensions. They do not establish that an artifact is a temporal “air ghost”: that claim would require correspondence to time-specific source measurements. The evidence instead narrows the immediate design target to support extent and first-surface competition, with source/pose errors still possible.

5.3. Oracle Ladder: Selection, Support and Geometry

For a fixed mesh and observed return, define H_0 as a correct first intersection; H_1 as any intersection within the range band; and H_2 as endpoint-to-surface distance within 0.2 m. Ignoring numerical tolerance, $H_0 \subseteq H_1 \subseteq H_2$. The first gap measures the opportunity for target-informed intersection selection; the second measures the lack of correct along-ray support despite Euclidean proximity. A final

observed-endpoint oracle would be 100% by construction and says nothing about hidden geometry.

For AdaPoinTr, the ladder is 14.29% $\rightarrow$ 15.43% $\rightarrow$ 84.60%. Selecting a rear intersection alone can recover little; native point completion plus surface conversion chiefly lacks the required along-ray support. For r4 it is 36.07% $\rightarrow$ 58.32% $\rightarrow$ 73.46%. Thus an existing correct intersection is hidden on 22.25 percentage points of rays. For r6 it becomes 39.07% $\rightarrow$ 55.03% $\rightarrow$ 72.44%. First-surface supervision improves literal use of support while reducing some of the available rear support. These values motivate separate support and placement mechanisms rather than treating all errors as one scalar-distance problem.

5.4. Pruning Is an Intervention, Not an Explanation

Removing all $q > 10$ faces changes none of these results. Removing components without build support lowers r4 early rate only from 30.80% to 29.89%, while raising misses from 20.39% to 26.27%. For AdaPoinTr, early improves from 8.54% to 4.27%, but misses rise from 54.24% to 80.21% and correct hits fall from 14.29% to 12.12%. Broad connectivity pruning therefore discards useful completion as well as errors.

We also remove the union of faces responsible for held-out early returns, then recast the original rays once. On r4, this oracle yields 48.76% hit, 12.39% early and 23.17% missing, compared with 36.07/30.80/20.39 before removal. Its hit rate remains below the per-ray any-hit oracle’s 58.32%: one shared mesh couples rays, deleting a front face can expose another early face, and a face can be useful to a different ray. This specific deletion policy is not the optimal global pruning oracle. Its result supports modeling physical support domains under positive-witness constraints, rather than implementing per-ray transparency.

5.5. Completed First-Surface and Joint Comparisons

Switching attraction to actual first-surface supervision reduces early by 7.815 pp (95% log interval $[-14.286, -2.820]$), but directed distance increases 0.01918 m ($[0.00247, 0.04363]$). Correct hits improve by 2.998 pp with an interval crossing zero. Relative to r3, r6 improves hit by 7.992 pp and reduces missing by 7.817 pp, while increasing early by 6.762 pp and free intrusion by 0.07264 m; these intervals exclude zero. r6 is an informative compromise, not a method that jointly wins on coverage and physical consistency.

The final joint r7 run completes 30 epochs and 11,130 updates. Against matched LiDAR r6, it improves correct first hits by 10.536 pp (95% log interval $[6.462, 14.621]$), with improvement in all five logs. Missing decreases 4.379 pp ($[-9.395, -0.292]$), and one-way distance decreases 0.07940 m ($[-0.18738, -0.00933]$). The joint path-

Method	Hit $\uparrow$	Early $\downarrow$	Miss $\downarrow$	Free $\downarrow$	Dist. $\downarrow$	Recall $\uparrow$
LiDAR PCA	21.24	4.35	73.41	0.018	0.305	65.42
VGGT native fusion	22.78	9.16	57.35	0.150	0.175	78.13
AdaPoinTr (task-tuned)	14.29	8.54	54.24	0.122	0.114	84.60
CAPA-style TTA	17.80	10.95	58.46	0.120	0.203	76.90
LiDAR R8	31.00	5.59	56.83	0.034	0.230	71.95
Open charts r3	31.08	16.22	28.37	0.165	0.179	69.58
Ray attraction r4	36.07	30.80	20.39	0.267	0.154	73.46
First surface r6	39.07	22.98	20.56	0.238	0.173	72.44
Joint r7	49.60	21.46	16.18	0.280	0.094	80.87

Table 1. Completed development results from canonical saved summaries. Hit, early, missing and recall are percentages; free intrusion and distance are meters. Rows span different model and adaptation protocols and are not all single-factor ablations. Directed distance is conditional on available/evaluable surfaces; empty cases remain in coverage and ray accounting. Native fusion can produce surfaces for five additional zero-LiDAR development entries, so conditional distance is not an unconditional comparison. r7/r6 is the final pathway comparison described in the text.

r6 minus r4	Difference	95% log interval
Hit (pp)	+2.9980	[-0.6018, +7.7197]
Early (pp)	-7.8152	[-14.2862, -2.8201]
Missing (pp)	+0.1699	[-1.5447, +1.8846]
Free (m)	-0.0287	[-0.0586, +0.0058]
Distance (m)	+0.0192	[+0.0025, +0.0436]
Recall (pp)	-1.0169	[-3.7172, +1.6730]

Table 2. r6 minus r4, five paired development logs. Early improves in every log; directed surface distance worsens in every log. Other intervals cross zero and do not establish preservation.

r7 minus r6	Difference	95% log interval
Hit (pp)	+10.5363	[+6.4624, +14.6212]
Early (pp)	-1.5225	[-5.7790, +2.7510]
Missing (pp)	-4.3793	[-9.3948, -0.2924]
Free (m)	+0.0420	[-0.0005, +0.0697]
Distance (m)	-0.0794	[-0.1874, -0.0093]
Recall (pp)	+8.4288	[-0.7072, +23.9799]

Table 3. Completed joint r7 minus LiDAR r6 on five paired development logs. Geometry/hit gains coexist with unresolved free-space preservation.

way therefore has supported development gains in observed geometry and literal hits under this representation.

Physical preservation is not established. Early changes -1.523 pp with interval $[-5.779, 2.751]$; mean free intrusion increases 0.04202 m with interval $[-0.00049, 0.06974]$, worsening in four logs. Recall changes $+8.429$ pp with interval $[-0.707, 23.980]$. These intervals do not prove non-inferiority. The final r7 values are 49.604% hit, 21.459% early, 16.179% missing, 0.27983 m free intrusion, 0.09392 m distance and 80.872% recall. This result supports retaining visual geometry as a useful pathway, while keeping free-space and surface-domain claims unresolved. The comparison changes native seeds, DPT features and build-depth auxiliary supervision together, with the same chart, physical objective and update

budget. It does not attribute the gain to a single head or feature level.

A subsequent fixed-surface r7 audit uses the same 75 actors and 11,886 owned returns, with zero inference or optimization. Its ladder is $49.60\% \rightarrow 61.97\% \rightarrow 80.87\%$; 12.36 percentage points still have an incorrect front surface with a correct rear intersection. Early triangles remain well-shaped. Build-supported components contribute 19.387 of the 21.459 early points, or 90.3%. Removing unsupported components raises missing from 16.18% to 25.11%. This latest evidence retains the domain/first-surface problem despite r7’s genuine development improvement.

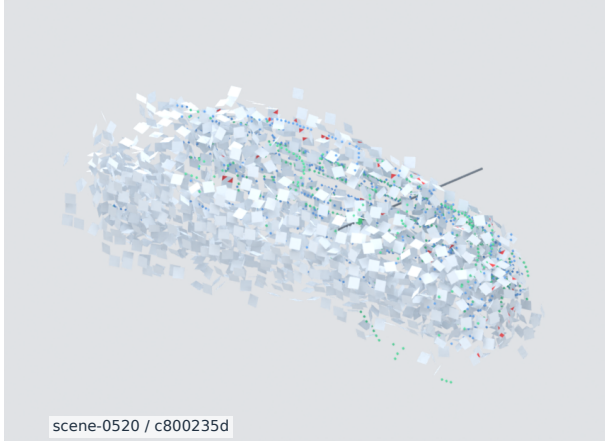
r7 requires 20,249.42 s (5.62 hours), including initial/final evaluation, with 10.224 GiB peak CUDA allocation and 34.127 GiB process RSS. It trains the 32.65M-parameter DPT and 1.71M-parameter query decoder. r6 takes 2,139.06 s with 0.210 GiB peak CUDA allocation. The comparison matches update count, not compute cost; these concurrent-system wall times are not exclusive GPU throughput.

5.6. New Operator: What Has Actually Been Verified?

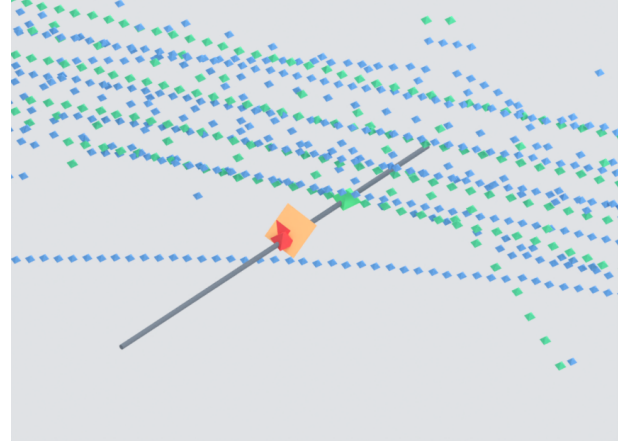
The new CPU float64 prototype is evaluated on 60 interior first-hit witnesses from one saved r6 development mesh with 1,024 vertices and 1,152 faces. This is a numerical operator check using known geometry, not training or generalization. The custom first-depth backward matches a central directional finite difference with relative error 6.68×10^{-10} . The sparse adjoint identity has absolute error 1.78×10^{-15} ; its custom backward matches the explicit scatter exactly. The true projected constraint residual is 2.01×10^{-7} , rather than the solver’s smaller internal residual estimate.

For displacement norm 10^{-3} m, maximum protected depth change is 8.87×10^{-9} m after projection versus 9.16×10^{-5} m for the normalized unprojected direction. Halv-

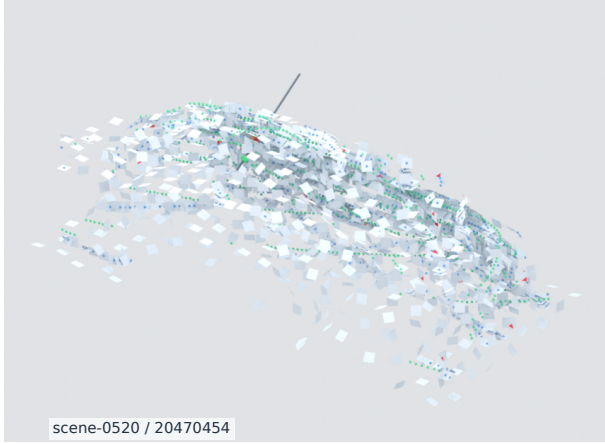
AdaPoinTr | canonical surface



AdaPoinTr | first-hit triangle: 0.274 m early



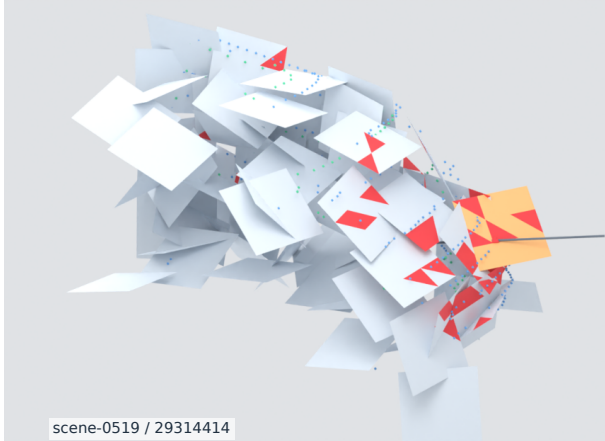
VGGT-native | canonical surface



VGGT-native | first-hit triangle: 0.323 m early



Joint-r7 | canonical surface



Joint-r7 | first-hit triangle: 0.339 m early

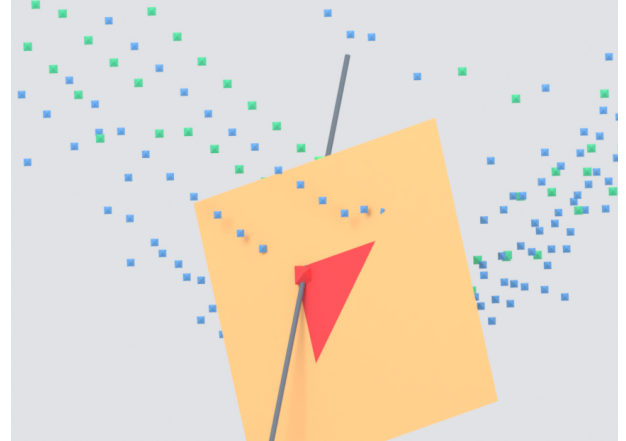


Figure 2. Blender renders of original saved surfaces. Rows show task-fine-tuned AdaPoinTr after PCA conversion, adapted VGGT native fusion, and final joint r7. Left shows actor context; right isolates the responsible connected chart using its original triangles. Gray is the original surface, red marks early-hit triangles, orange highlights the selected chart, blue markers are build points and green markers are heldout endpoints. The black segment is the original ray. Isolation affects display only; all queries use the full saved surface. Actors are selected by median early rate among predefined eligible failures, not by the largest improvement.

Exact intersection of original triangles with a plane containing the selected sensor ray

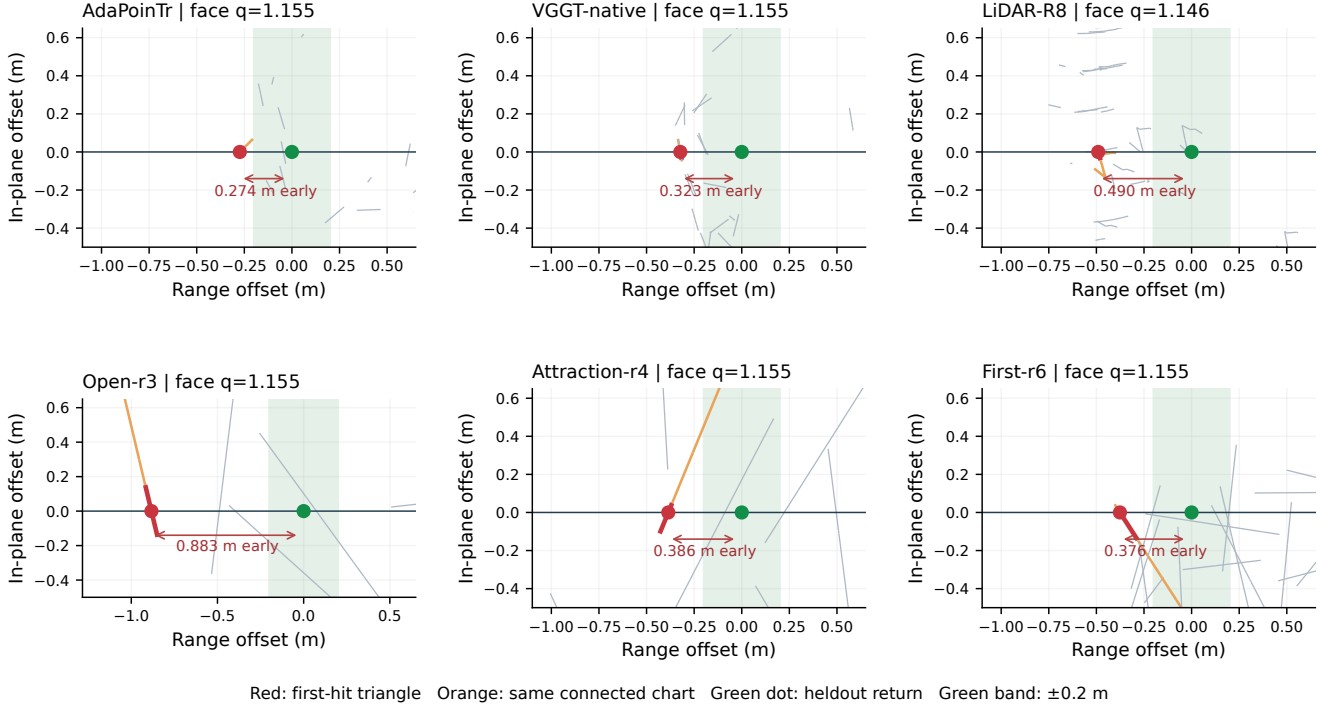


Figure 3. Exact geometric failure slices for the six independently selected examples. The plane contains the original sensor ray; plotted segments are triangle-plane intersections, not projected point-cloud silhouettes. Red identifies the actual first-hit triangle and orange its connected chart. The green dot is the measured return and the band is the ± 0.2 m tolerance. Regular local triangles can produce a wrong front crossing even when another surface is correctly supported behind it. Different panels show different actors and are morphology illustrations, not matched visual ablations.

ing the projected step twice reduces the change by approximately four each time, consistent with the local second-order statement. No finite-step protection of all rays is inferred. Autograd records 5,820 bytes of selected-hit tensors for this case, rather than an all-ray/all-face tape. The complete check takes about 0.10 s and process peak RSS is 0.643 GiB. These are CPU measurements; CUDA integration, end-to-end training cost and learned gains remain unmeasured.

6. Limitations and Research Decisions

The defensible output is a partial rigid surface under known trajectories. Current results do not support claims about complete hidden shape, articulated actors, new sensor intensity/dropout, image realism, learned motion, or closed-loop simulation. Background disocclusion and full-scene ownership remain separate problems. Unknown space stays unknown. Coordinate-equivariance checks and editable demos are interface evidence, not counterfactual ground truth.

The chart formulation, new operator and proposed domain constraints are in the method section because they

define the intended solution. Their evidence levels differ. Canonical charts and first-surface supervision are implemented and trained; the sparse orthogonal operator is implemented and numerically checked; learned chart boundaries and end-to-end witness-protected training are proposed. The present evidence does not establish their novelty against all prior work or their empirical superiority. A credible method claim needs a matched trained comparison, support-retention tests, and fixed external confirmation.

We prune three immediate stories: sliver regularization as the explanation for the inspected early surfaces; indiscriminate component pruning as a safe fix; and a larger visual backbone as sufficient evidence of method novelty. A targeted domain boundary with shared first-hit witnesses is the remaining mechanism to test. Merely adding more free-space loss is also insufficient when a ray lacks correct support. This decision follows the observed error geometry rather than an open-ended loss or backbone search.

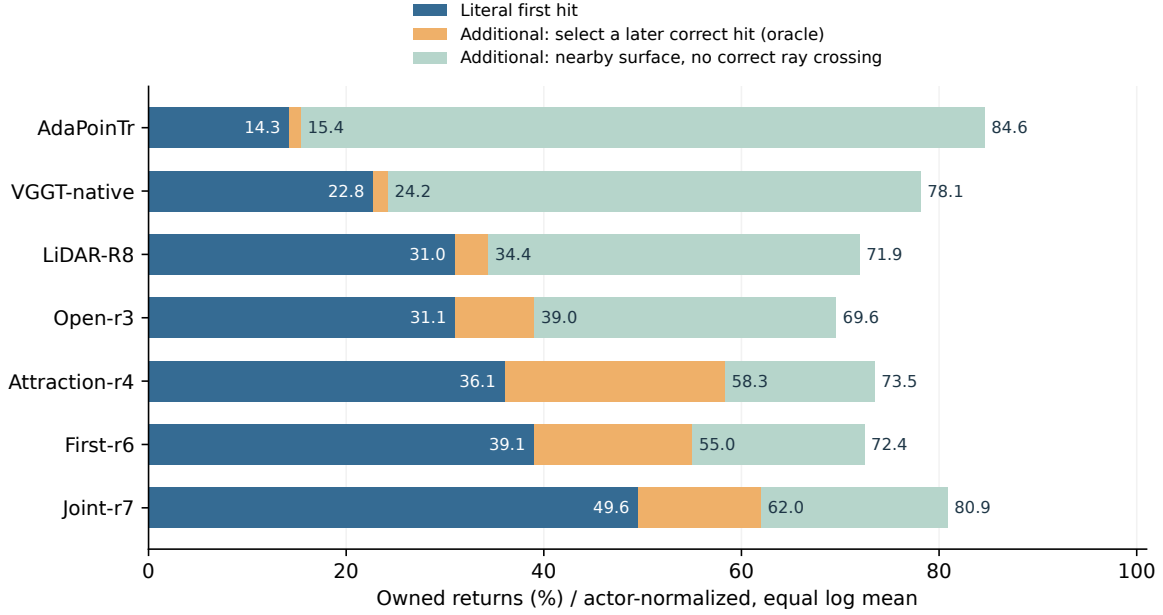


Figure 4. Oracle ladder on the same saved surfaces and owned returns. Blue is literal first-hit accuracy; orange adds rays with a later correct intersection; green adds rays whose endpoint is near a surface but has no correct intersection. The last two levels use the heldout endpoint and are diagnostic relaxations. Near-surface coverage is not a realizable ray predictor or a dense-surface ground-truth score.

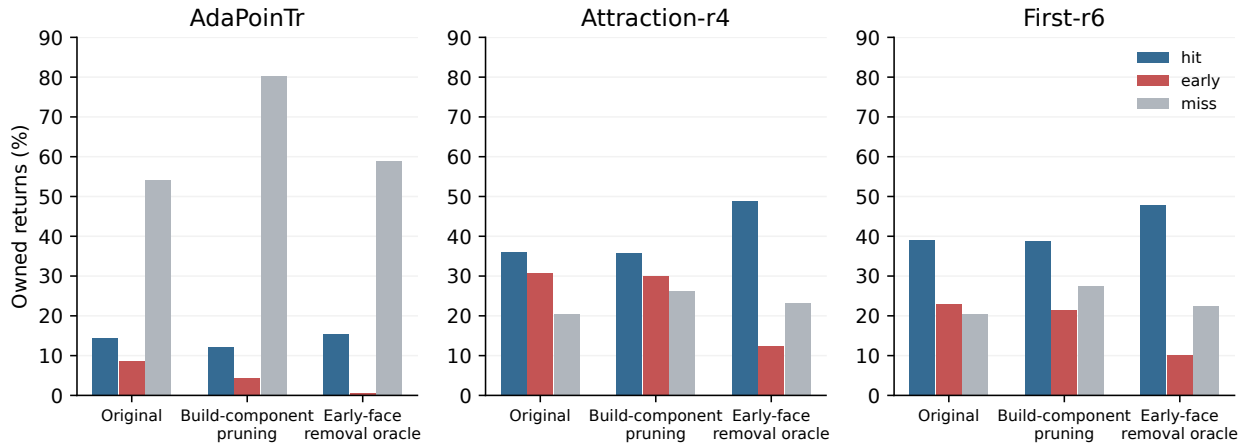


Figure 5. Recast after shared-mesh interventions. Build-component pruning uses only build support. The early-face removal oracle uses the union of heldout early-hit faces, removes them once, and recasts every owned ray against the same remaining mesh. It does not pick a different surface for each ray. Neither intervention is a deployed result.

7. Conclusion

Queryable canonical surface reconstruction gives this project a bounded geometric object and an independent evaluation target. Exact failure surfaces show why good local triangles and close endpoints do not guarantee correct first intersections. The oracle decomposition motivates a support-domain representation and incidence-normalized witness constraints; a sparse custom operator makes that proposal concrete and testable. Current training reveals a real coverage/first-surface trade-off, while the proposed or-

thogonal and boundary mechanisms remain to be validated as a learned method.

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A. Method Claims and Evidence Levels

This is a research-development manuscript. The following matrix prevents the completed V7.3 system from being conflated with the new method proposal.

What would make this a strong method claim? The central hypothesis is that *separating support boundaries from embedding, while preserving shared first-hit witnesses, removes unsupported chart extensions without destroying useful completion*. The relevant experiment is a fixed-observation, fixed-embedding-capacity comparison of ordinary charts, orthogonal updates alone, domains alone, and their joint model. It should include literal hard queries, endpoint coverage, miss accounting, free intrusion, domain area and retained positive witnesses. Model selection remains on development data, followed by one fixed external confirmation. The current task does not launch this new training grid or modify r7.

Why this is more specific than another head. The domain changes the set of geometric points that exist, and therefore all queries of the exported surface. The witness constraint couples different observations through shared vertices; it cannot be reproduced by a scalar confidence on each ray. The sparse adjoint implements derivatives of that shared geometric query. These are the representational and mathematical commitments of the proposal. However, a learnable b_k head with no support preservation and no matched evidence would still be a weak novelty claim.

B. Derivation and Limits of the Orthogonal Constraint

Let $e_1 = v_1 - v_0$, $e_2 = v_2 - v_0$. The selected intersection solves

$$[d, -e_1, -e_2] \begin{bmatrix} t \\ u \\ v \end{bmatrix} = v_0 - o, \quad \beta = (1 - u - v, u, v). \quad (15)$$

Nonzero determinant is equivalent to nonzero incidence. The differential of $o + td = \sum_j \beta_j v_j$ satisfies

$$d\delta t = \sum_j \beta_j \delta v_j + \sum_j v_j \delta \beta_j, \quad \sum_j \delta \beta_j = 0. \quad (16)$$

The second sum lies in the original triangle’s tangent plane. Multiplying by $n^\top$ gives Eq. 9. No derivative of an assumed constant barycentric point was substituted for this argument. The barycentric changes disappear because the tangent-plane term is orthogonal to n .

For an SVD $B = U\Sigma V^\top$, $B^\dagger B$ projects onto the span of right singular vectors with nonzero singular values. Con-

sequently

$$B(I - B^\dagger B) = 0, \quad \langle (I - B^\dagger B)g, B^\dagger c \rangle = 0. \quad (17)$$

When B has dependent rows, $BB^\dagger c$ is only the projection of the desired depth correction onto the feasible constraint space. The residual $c - BB^\dagger c$ quantifies conflicts. Damping a normal-equation solve makes the protection approximate and must not be described as exact orthogonality. The prototype instead applies LSMR to $B^\top$ through sparse operators, checks the *true* residual, and does not use the recursively reported solver residual as proof.

The sparse representation has three vertex indices and nine float coefficients per valid witness. The forward query first selects the actual front face using geometry alone; target range never chooses that face. For misses, the custom depth function returns infinity and does not manufacture gradients. A training loss must call the existing constructive branch on misses. Rays and calibration are fixed in this implementation; their derivatives are not returned.

Visibility remains discrete. Moving a triangle across a ray can change a miss to a hit discontinuously; two triangles can exchange front order. Normalization cannot repair either event. An orthogonal step also protects no unlisted ray, and a global neural weight update can affect other actors. A future trainable version may project the decoder’s output update and distill it into the network, or use Jacobian-vector products through the decoder, but these are different algorithms with different guarantees. They must not be silently identified with the vertex-space prototype.

C. A Concrete Domain-Clipping Operator

The proposed boundary field b_k is piecewise linear on a shared intrinsic chart grid. On an edge from u_a to u_b with opposite endpoint signs $s_a = b(u_a)$ and $s_b = b(u_b)$, its zero crossing is

$$u_* = (1 - \lambda)u_a + \lambda u_b, \quad \lambda = \frac{s_a}{s_a - s_b}. \quad (18)$$

Within a fixed sign pattern,

$$\frac{\partial \lambda}{\partial s_a} = \frac{-s_b}{(s_a - s_b)^2}, \quad \frac{\partial \lambda}{\partial s_b} = \frac{s_a}{(s_a - s_b)^2}. \quad (19)$$

Each shared edge emits one crossing vertex, reused by adjacent clipped triangles. Interpolation of the piecewise-linear embedding produces the physical crossing. A face emits a triangle or quadrilateral, with a deterministic triangulation used identically in training and evaluation. Storing each edge crossing once closes geometry bookkeeping and bounds storage by the chart grid size. There is no positive volume thickness or ray-specific mask.

Component	Present evidence	Missing evidence / attribution limit
Canonical reusable actor charts	Implemented; exported metric meshes; shared closest-point and hard-ray readout	Canonical coordinates and charts are established ideas; task formulation alone is not empirical novelty.
Trainable native geometry pathway	Full DPT and query training; r7 improves development hit/miss/distance against r6	Native seeds, features and auxiliary loss change together. Free-space preservation and external confirmation remain unresolved.
Actual first-surface loss with constructive miss branch	Trained r6/r7; r6-r4 paired decrease in early with worse distance	A useful geometric objective, but not evidence of a new representation by itself. Silhouette and grazing behavior remain.
Witness-constrained chart domain	Formal representation, constraints and edge derivative in this manuscript	Boundary prediction, consistent clipped topology, observation conflicts and training are not implemented/evaluated.
Incidence-normalized orthogonal update	Derived; sparse gather/scatter custom adjoint; actual-mesh numerical test	Not in r7; no end-to-end optimization or learned improvement. Fixed-visibility guarantee only.
Sparse first-depth custom backward	Implemented BVH selection and selected-face adjoint; finite-difference and saved-tensor measurements	CPU implementation; no CUDA throughput or global visibility derivative claim.

Table 4. Method-level inventory. “Proposed” is a substantive specification, not an implemented result. None of these labels implies publication acceptance or overall SOTA performance.

This is an operator specification, not a completed clipping implementation. Opposite signs close to zero can create unstable derivatives; signs changing across an entire cell are a discrete topology event. A chart initialized with no positive domain receives no generated-surface attraction through an empty mesh. Keeping positive build witnesses and a nonempty coarse support is therefore essential. Reusing the old free objective without such constraints would recreate the established support-collapse failure V73-F02. Domain learning must also preserve the original metric and physical readout; it cannot improve the paper by merely thresholding away all difficult outputs.

D. Complete Oracle and Pruning Definitions

The first query uses Open3D RaycastingScene’s actual first intersection. All intersections and unsigned distance provide the two oracle relaxations [14]. These open surfaces do not justify signed-distance occupancy tests. The analysis reads original stored vertices/faces, original heldout origins/directions/ranges, and known positive-actor membership. No neural inference or optimizer update is performed. The six-method audit takes 72.70 s on CPU. Its initial empty-frame I/O error was corrected before the complete result; no cases were discarded.

Selection and rendering. For each method, eligible examples have at least 20 owned returns and five early returns. We select the actor whose early fraction is closest to the median among eligible actors, breaking ties by owner ID. Within that actor, an early ray with a later correct crossing is selected at median early severity when such rays exist.

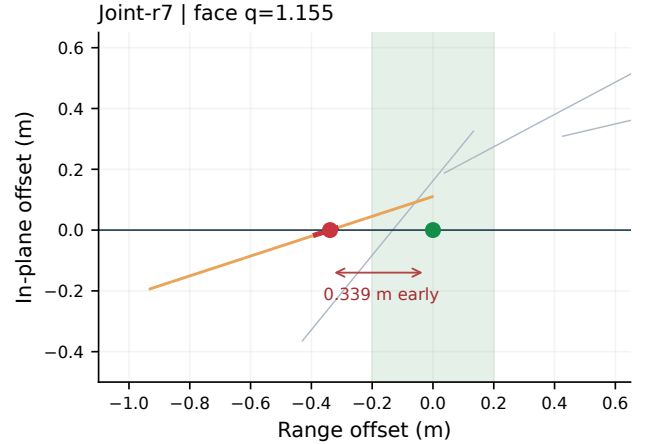


Figure 6. Exact slice of the completed r7 example in Fig. 2. The responsible regular triangle crosses 0.339 m before the measured return. Gray shows other original triangles; the isolated chart in the Blender detail is orange here. The query uses this full surface.

This intentionally visualizes a diagnosed failure class, not representative population accuracy. Raw actor counts and selection details are retained. No comparison panel is presented as a same-actor improvement unless explicitly stated.

The whole-actor Blender scene contains original surface vertices and faces. Auxiliary octahedra mark observations and cylinders mark rays; they are visualization objects only. The detail scene isolates the selected connected chart and nearby measurement markers to expose the chosen first-hit face, with no geometric smoothing, remeshing, or generated image content. Other charts are hidden only in that

Method	First $\uparrow$	Any $\uparrow$	Near $\uparrow$	Early $\downarrow$	Unsupported early
AdaPoinTr	14.29	15.43	84.60	8.54	4.32
VGGT native	22.78	24.23	78.13	9.16	4.43
LiDAR-R8	31.00	34.38	71.95	5.59	0.01
Open-r3	31.08	39.04	69.58	16.22	1.64
Attraction-r4	36.07	58.32	73.46	30.80	1.63
First-r6	39.07	55.03	72.44	22.98	2.06
Joint-r7	49.60	61.97	80.87	21.46	2.07

Table 5. Exact oracle ladder and unsupported-component early contribution, all in percentage points of owned returns. “Unsupported early” is a subset of early under the stated build-component test. It is not a classifier of physical floaters.

display. The analytic slice separately intersects every original triangle with a plane through the selected ray. This gives a checkable geometric view of the surrounding surface as well as the isolated responsibility chart.

Component support is computed as the minimum unsigned distance from any original build endpoint to that component’s triangles. A 0.2m threshold matches the surface-recall tolerance and is not tuned per method. The shape threshold is fixed at $q > 10$. Both are descriptive tests. Build-component pruning deletes all unsupported components once. The early-face oracle deletes the union of triangles selected as first early hits over owned heldout rays, then casts the same rays once against the remaining shared surface. A new front face can still be early; the oracle is not iterated to erase all violations.

E. Historical Results Retained as Context

The completed V7.3 history already includes closed shared meshes, a native-only adapted DPT, short-window and full-track supervision, narrow patches, a beam free-space objective, an event objective, CAPA-style adaptation, AdaPoinTr task fine-tuning, Actor TSDF, and background composition. These comparisons establish useful failure boundaries, but combining them into one narrative does not make each an independent method contribution.

Earlier closed Q-v2 joint r1 has 7.454pp more early returns than its same-grid LiDAR r2, with interval [1.122, 17.719]; the other five paired intervals cross zero. Its 67 nonempty meshes have no detected nonadjacent self-intersection, whereas the LiDAR control has such intersections in 48 of 67. Thus closed-joint failure was not directly explained by slivers or detected self-intersection. Opening the charts also did not establish joint coverage/physics dominance. These observations justify keeping topology, support direction, and visibility explanations separate.

AdaPoinTr uses its full 32,494,657-parameter architecture, 512 queries and 16,384 output points, initialized from the official PCN checkpoint and task-fine-tuned for 30 epochs. The task interface swaps the project’s Z-up coordinates to PCN Y-up and back, using known actor size for

isotropic scaling. Its common-budget physical output selects centers by FPS and constructs PCA patches with fixed spacing. This conversion is part of the evaluated pipeline. A saved density diagnostic that instead uses all output points raises hit by 24.72 pp and early by 30.85 pp, with 0.2190 m more free intrusion. More rendered support is not unconditionally better.

Static background composition can dominate full-scene errors and must be isolated. Existing development background TSDF/carving reduces free intrusion while increasing missing returns. The actor task does not turn an unobserved region uncovered by an edit into known background truth. Trajectory edits provide an interface demo only. External confirmation uses Argoverse 2 [13]; it is cross-dataset, not an i.i.d. nuScenes test.

F. Reproducibility and Remaining Evaluation

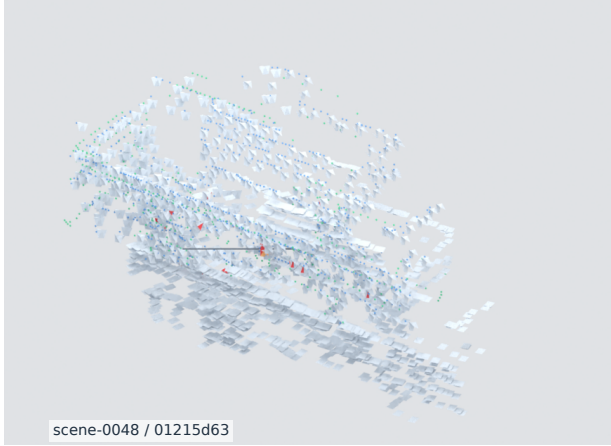
The source branch is `research/worldsim-v7.3-geometric-adaptation`. The completed r7 training code is 185bc728; its full DEV summary was generated once after completion and its ledger was committed as 6499d34d. Run identities are retained, without extra checksums or fingerprints. The two primary runs are:

- r6: `WS-V73-Q-V2-01/20260909T123000Z_open-charts-lidar-first-surface-s7304-r6`.
- r7: `WS-V73-Q-V2-01/20260909T131000Z_open-charts-joint-first-surface-s7304-r7`.

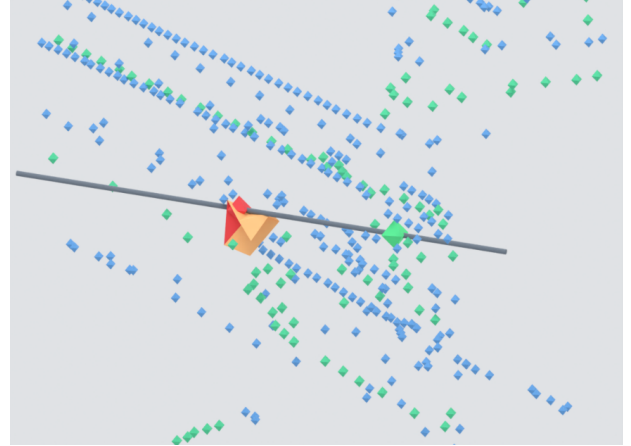
All runs are beneath `/root/autodl-tmp/runs/worldsim_v73/`. The paired DEV evidence is `docs/autoresearch/worldsim_v73/ray_support/first_surface_r6_analysis.json` and `open_joint_r7_analysis.json` in that same evidence directory. The six-method forensics is stored under `docs/autoresearch/worldsim_v73/paper_forensics/20260909T184200Z__saved-surface-oracles-r1/`; the r7 extension has its own run so the original six-method audit remains identifiable.

The operator check is `operator_check.json` in-

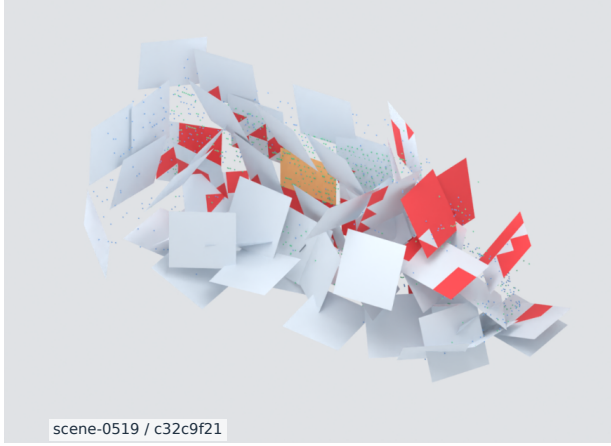
LiDAR-R8 | canonical surface



LiDAR-R8 | first-hit triangle: 0.490 m early



Open-r3 | canonical surface



Open-r3 | first-hit triangle: 0.883 m early

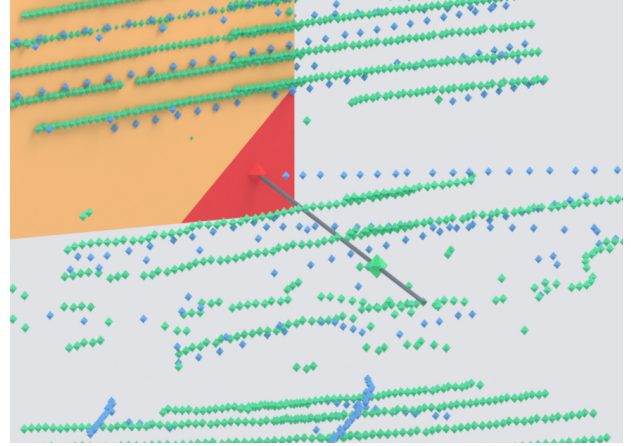


Figure 7. Additional saved-surface renders for the LiDAR-R8 and open-chart r3 controls. The left image preserves whole-actor context and the right exposes the selected early-hit triangle. Colors and independent failure selection follow Fig. 2. All query measurements use the full original surface.

side the first forensic directory. Scripts are `scripts/forensic_worldsim_v73_paper_surfaces.py`, `scripts/forensic_worldsim_v73_paper_r7.py`, `scripts/render_worldsim_v73_paper_blender.py`, and `scripts/worldsim_v73_witness_operator_prototype.py`. The manuscript, derived plots, selected raw cases and Blender scenes are delivered together. The official Blender color-management assets must be available when rerendering.

Twenty fixed AV2 logs contain 936 actor entries, including 878 build-ready and 58 missing-input entries. Their construction was completed before quality evaluation. The original research task launched the preregistered fixed r7/r6 confirmation after DEV completion; this manuscript does not substitute its own selection or alter that run. No completed external quality result was available at this revision’s evidence cutoff. External results, full-scene composition,

new support-domain training and end-to-end orthogonal updates must not be inferred from the local forensics or numerical prototype.

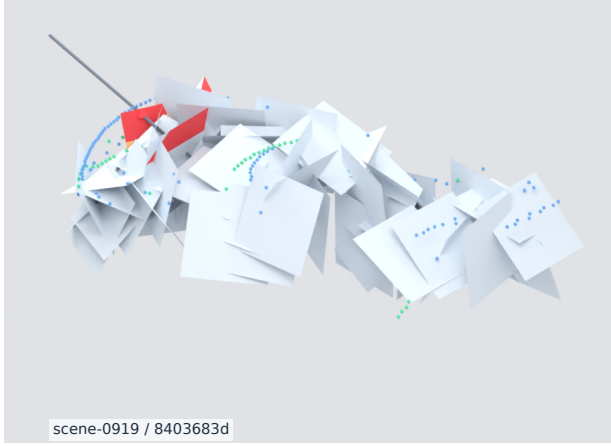
Attraction-r4 | canonical surface



Attraction-r4 | first-hit triangle: 0.386 m early



First-r6 | canonical surface



First-r6 | first-hit triangle: 0.376 m early

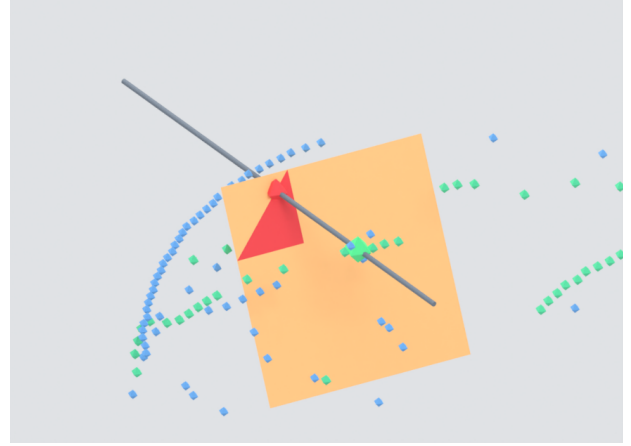


Figure 8. Additional saved-surface renders for attraction r4 and first-surface r6. These are independently selected actors, not a matched before/after comparison. A regular chart can be locally supported while extending into the selected ray before its observed return.

Method	Hit (%)	Early (%)	Miss (%)	Free (m)	Recall (%)
LiDAR PCA	21.24	4.35	73.41	0.0177	65.42
Actor TSDF + carving	3.28	0.26	96.33	0.0008	21.51
Native fusion	22.78	9.16	57.35	0.1503	78.13
LiDAR R6 (window labels)	31.01	14.02	49.31	0.0838	72.42
LiDAR R7 (track labels)	31.75	11.84	49.90	0.0716	74.55
LiDAR R8 (beam free)	31.00	5.59	56.83	0.0343	71.95
LiDAR R9 (+event)	32.47	6.01	54.57	0.0385	72.47
CAPA-style TTA R2	17.80	10.95	58.46	0.1203	76.90
AdaPoinTr R2	14.29	8.54	54.24	0.1221	84.60
Joint R5 (window labels)	22.28	20.98	37.71	0.3517	79.71
Native DPT R11 (track labels)	17.61	7.64	60.09	0.1650	75.15
Joint R10 (track, hard free)	27.15	20.26	32.62	0.2710	82.60
Joint R12 (track, beam free)	21.59	5.64	65.15	0.0606	76.22
Native R14 (track, beam free)	20.56	9.99	57.52	0.1664	77.20
Q-v2 joint r1	28.46	26.94	30.22	0.1793	59.48
Q-v2 LiDAR r2	30.55	19.49	25.54	0.1405	64.82
Open charts r3 (LiDAR)	31.08	16.22	28.37	0.1652	69.58
Ray attraction r4 (LiDAR)	36.07	30.80	20.39	0.2665	73.46

Table 6. Previously completed V7.3 methods retained from the interim r4 archive. Different rows have different labels, adaptation and surface-conversion protocols. This table supplies history; the main paired conclusions use r6–r4 and r7–r6. The latest two methods appear in Table 1.